

# Analysis

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May 11, 2026

**Theorem 1** (Rolle's Theorem).

**Theorem 2** (Mean Value Theorem). Let  $f : \mathbb{R} \rightarrow \mathbb{R}$  be a function that is continuous on  $[a, b]$  and differentiable on  $(a, b)$ . Then there exists a  $c \in (a, b)$  such that

$$f'(c) = \frac{f(b) - f(a)}{b - a}$$

Proof: Consider the function  $g(x) = f(x) - \frac{f(b)-f(a)}{b-a}(x-a)$ . Then  $g(a) = f(a)$  and  $g(b) = f(a)$ . Thus by Rolle's Theorem there exists a  $c \in (a, b)$  s.t.  $g'(c) = 0$  and the rest follows.  $\square$

**Theorem 3** (Taylor's Theorem). Let  $f : \mathbb{R} \rightarrow \mathbb{R}$  be a function that has its first  $n$  derivatives continuous on  $[a, b]$  and differentiable on  $(a, b)$ . Fix a point to expand around,  $x_0$ . Let  $x \in [a, b]$ . Then for any  $n \in \mathbb{N}$  and any  $c \in (a, b)$  there exists and  $x_0 \in (a, c)$  such that

$$f(c) = \sum_{k=0}^n \frac{f^{(k)}(x_0)}{k!} (x - x_0)^k + \frac{f^{(n+1)}(c)}{(n+1)!} (x - x_0)^{n+1}$$

Idea of Proof:

- Construct a function  $F(t)$  such that
  1.  $F$  is continuous on  $(a, b)$  and differentiable on  $[a, b]$ .
  2.  $F(x) = f(x)$  and  $F(x_0) = f(x_0)$
  3.  $F'(t) = 0$  forces the correct coefficient of  $(x - x_0)^{n+1}$
- Use the mean value theorem on  $F(t)$ .

Noteworthy property of Modified Taylor Series: If  $P(t) = \sum_{k=0}^n \frac{f^{(k)}(t)}{k!} (x - t)^k$  then

$$P'(t) = \frac{f^{(n+1)}(t)}{n!} (x - t)^n$$

Proof:( following Lay 28.2) Let  $x \in [a, b]$  and fix  $n \in \mathbb{N}$ . Let

$$F(t) = \sum_{k=0}^n \frac{f^{(k)}(t)}{k!} (x - t)^k + M(x - t)^{n+1}$$

where  $M$  is the unique solution to

$$f(x) = \sum_{k=0}^n \frac{f^{(k)}(x_0)}{k!} (x - x_0)^k + M(x - x_0)^{n+1}.$$

Note that by the noteworthy property mentioned above we have

$$F'(t) = \frac{f^{(n+1)}(t)}{n!} (x - t)^n + (n + 1)M(x - t)^n.$$

Note also that  $F(x) = F(x_0) = f(x)$ , so invoking the mean value theorem (or Rolle's Theorem) we obtain that there exists a  $c$  between  $x$  and  $x_0$  strictly such that

$$F'(t) = 0 = \frac{f^{(n+1)}(t)}{n!} (x - t)^n + (n + 1)M(x - t)^n.$$

Solving for  $M$  gives  $M = \frac{f^{(n+1)}(x_0)}{(n+1)!}$  as required.  $\square$